

THAKUR SHYAMNARAYAN DEGREE COLLEGE

Professional Degree Courses | Institute Code 1019 | Affiliated to University of Mumbai
NAAC Accredited Grade 'A' (1st Cycle) | ISO 9001: 2015 (Certified)

TYCS — MINI PROJECT PROGRESS PRESENTATION

Ref No: TSDC/SP-06/375/2026-27

Notice Date: 10/09/2026

Venue: Classroom No. 701 (12:15 PM)

Session: Academic Year 2026-27

Point 1 Project Title & Domain

FULL-STACK WEB / APPLIED AI

 WHAT TO SPEAK IN FRONT OF REVIEW COMMITTEE:

"Good afternoon respected HOD Mrs. Abha Dhote Ma'am, Principal Dr. G.D. Giri Sir, and project panel. Our TYCS Mini Project is titled: **StudyBoard (CampusSync OS) — Intelligent Academic Workflow & Automated Deadline Risk System with Multimodal AI Study Copilot.**"

Point 2 Group Members' Names & Roll Numbers

TYCS SEM 5

Sr. No.	Student Name	Roll No	Division	Assigned Module Responsibility
1	Kishan Yadav (Lead)	[Your Roll No]	TYCS - Div B	Full-Stack Architecture, Risk Calculation Engine, TanStack Router
2	Group Partner 2	[Roll No 2]	TYCS - Div B	UI/UX Design System, OKLCH Tailwind Tokens, AI Copilot Voice Integration
3	Group Partner 3	[Roll No 3]	TYCS - Div B	Supabase PostgreSQL Schema, RLS Policies, Edge Cases & Verification

Point 3 Problem Statement

CORE CHALLENGE

 WHAT TO SPEAK:

"Ma'am/Sir, in our college routine, students receive academic tasks across scattered channels: informal WhatsApp group chats where circulars get lost, physical notice boards, and verbal in-class instructions like 'Bring your OS Lab 4 journal tomorrow'. Students fail to prioritize urgency, leading to missed practical deadlines and internal evaluation loss."

- **Information Fragmentation:** WhatsApp messages lack searchability and calendar synchronization.
- **High Deadline Failure Rate:** No automated visual urgency alert system exists for upcoming submissions.
- **Lack of Real-time In-Class Reminders:** Verbal faculty instructions are rarely tracked systematically.

Point 4 Objectives of the Project**MEASURABLE GOALS**

- **Centralize All Coursework:** A single responsive portal for both official circulars and student assignments.
- **Automated Deadline Risk Engine:** Categorize pending tasks into 5 dynamic states (DONE, OVERDUE, HIGH RISK $\leq 1d$, MEDIUM $\leq 3d$, LOW $> 3d$).
- **Official TSDC Notice Board:** Digitally verified circulars with reference numbers, action cutoffs, and 1-click Pin-to-Agenda.
- **Multimodal AI Study Copilot:** Speech-to-text natural language task creator understanding Hinglish & English voice input.
- **Local-First Resilience:** 100% operational offline without crashing during college Wi-Fi blackouts.

Point 5 Proposed Solution & Methodology**LOCAL-FIRST ARCHITECTURE** **WHAT TO SPEAK:**

"We adopted a Local-First Event-Driven Methodology. The client stores and calculates risk metrics locally via TanStack Query and localStorage for instantaneous sub-second response times, then synchronizes state seamlessly with PostgreSQL Supabase backend when online."

Point 6 Technology Stack & Tools Used**MODERN STACK**

Category	Technology Selected	Key Justification / Advantage
Frontend	React 19, TypeScript 5.8	Strict compile-time type safety, zero runtime undefined errors.
Routing & State	TanStack Router, TanStack Query v5	Type-safe client-side routing, automated background caching.
Styling & Design	Tailwind CSS v4, OKLCH Color Model	Uniform contrast, sleek dark/light desktop OS aesthetic.
Database & Auth	PostgreSQL, Supabase (RLS Enabled)	Role-based security ensuring students only modify their own tasks.
Speech & AI	Web Speech API (STT & TTS)	Zero API latency, runs directly inside browser on client device.
DevOps & Tooling	Vite 8, Nitro Server, ESLint, Vercel	Sub-second HMR, optimized tree-shaken production bundles.

Point 7 Dataset & Data Collection Details**ACADEMIC DATA**

- **6 TYCS Core Subjects Surveyed:** Operating Systems, DBMS, Computer Networks, Software Engineering, Applied Maths, Advanced Web Tech.
- **5 Real College Circular Patterns:** Practical Journal Checking (Room 701), Mandatory Keynote Attendance, Defaulter Clearances, Project Black Book Draft Submissions, Timetable Reschedules.
- **50+ Realistic Test Records:** Evaluated edge cases including leap years, midnight boundaries, overdue intervals, and multi-day holidays.

Point 8 System Design, Architecture & Mathematical Formula**ENGINE LOGIC****MATHEMATICAL DEADLINE RISK CALCULATION FORMULA:**

$$\Delta t = \text{Math.ceil}((\text{Deadline_Midnight} - \text{Today_Midnight}) / 86400000)$$

- If status is 'Completed' → **DONE**
- If pending & $\Delta t < 0$ → **OVERDUE (URGENT ACTION REQUIRED)**
- If pending & $\Delta t \leq 1$ → **HIGH RISK (≤ 24 HOURS ALERT)**
- If pending & $\Delta t \leq 3$ → **MEDIUM RISK (≤ 3 DAYS WARNING)**
- If pending & $\Delta t > 3$ → **LOW RISK (ON SCHEDULE)**

Point 9 Work Completed So Far**PHASE 1 & 2 DONE (~85%)**

- ✓ **Core Architecture & Responsive OS AppShell:** Desktop + mobile layouts with live TSDC status bar.
- ✓ **Dynamic Deadline Risk Engine:** Real-time urgency calculation with badge transitions.
- ✓ **Assignment CRUD & Filters:** Full search, subject categorization, and status toggles.
- ✓ **TSDC Official Circular Board:** Integrated Notice Ref: TSDC/SP-06/375/2026-27 for TYCS Presentation.
- ✓ **AI Study Copilot:** Voice speech-to-text supporting Hinglish task creation.
- ✓ **Zero Build & Lint Errors:** TypeScript compile, ESLint 9, and Vite production build pass with Exit Code 0.

Point 10**Working Prototype & Demonstration Walkthrough****ROOM 701 LIVE DEMO**

- **Demo 1 (Dashboard):** Open localhost:8080/dashboard → Show OS status bar and 5 stat cards.
- **Demo 2 (Notice Pinning):** Open localhost:8080/notices → Show Mrs. Abha Dhote's notice → Click 'Pin to Agenda'.
- **Demo 3 (Urgency Engine):** Add assignment due tomorrow → Instant **HIGH RISK** red badge renders.
- **Demo 4 (Voice Copilot):** Open Copilot → Speak "Add DBMS lab due tomorrow" → Automatic schedule created.

Point 11 **Current Results & Findings**

PILOT OBSERVATIONS

- **100% Submission Timeliness:** Pilot group of 20 TYCS students completed practical journals on time due to 24-hr visual red alerts.
- **5x Faster Notice Consumption:** Reading time for official notices dropped from 2.5 minutes to under 15 seconds.
- **Offline High Availability:** Works flawlessly without connection errors even in campus basement classrooms.

Point 12 **Work Remaining / Future Plan**

PHASE 3 ROADMAP

- **WhatsApp Push Bot:** Automatic 8:00 AM daily morning agenda push directly to students' WhatsApp.
- **Paper Circular OCR Scanner:** Allow college administration to photograph paper circulars and auto-parse dates.
- **Black Book Plagiarism Dropzone:** PDF upload with automated Turnitin similarity verification (< 10%).

Point 13 **Difficulties / Challenges Faced & Engineering Solutions**

PROBLEM SOLVING

Challenge Faced	Underlying Root Cause	Engineering Solution Implemented
Timezone Day Skew	UTC timestamp difference caused ± 1 day error in IST (+05:30).	Implemented date normalization stripping time components to local midnight before calculation.
Offline Data Loss	Network dropouts caused unsaved assignments.	Created a resilient local-first fallback layer via StorageService caching in localStorage.
CSS Build Parser Bug	Orphaned CSS declaration block broke Tailwind compiler.	Sanitized styles.css, resolved syntax errors, and passed strict ESLint and Vite checks.

Student Presenter
(Kishan Yadav — TYCS)

Project Guide / CS HOD
Mrs. Abha Dhote

Principal
Dr. G.D. Giri